Youssef Menisy

Cairo, Egypt +20 155 380 1197 youssefmenisy2004@gmail.com
linkedin.com/in/Menisy2004 github.com/Menisy2004

Professional Summary

Results-driven AI Engineer with a strong Computer Science foundation from E-JUST, experienced in building scalable Generative AI systems, RAG pipelines, and agentic workflows. Proficient in orchestrating LLM solutions using LangChain, LangGraph, and FastAPI, integrating persistent memory (mem0), vector databases (pgvector), and cloud LLM APIs. Brings end-to-end deployment discipline with Docker containerization, CI/CD, and system observability (Langfuse, Prometheus, Grafana). Seeking to deliver robust, enterprise-grade AI automation across business lines.

Education

Egypt-Japan University of Science and Technology (E-JUST) **Sep. 2022 – May 2027**
Bachelor of Engineering in Computer Science *Alexandria, Egypt*

Technical Skills

Generative AI: LLM applications, RAG, prompt engineering, LangChain, LangGraph, retrieval workflows, semantic search, FAISS, pgvector, Sentence Transformers, mem0
Programming and APIs: Python, SQL, Java, C, FastAPI, REST APIs, asynchronous APIs, Server-Sent Events (SSE)
Machine Learning: PyTorch, scikit-learn, model evaluation, feature engineering, classification, neural networks, Random Forest, SVM
Data and Computer Vision: Pandas, NumPy, OpenCV, YOLO, Segment Anything Model (SAM), CUDA, ETL, data pipelines
Deployment and Observability: Docker, Git, GitHub, Prometheus, Grafana, Langfuse, Redis/Valkey, Jupyter Notebook
Languages: Arabic (Native), English (Fluent), Spanish (Beginner)

Selected Projects

AI Question Generator | *GitHub*
Python, FastAPI, LangChain, Gemini API, FAISS, Ollama
- Built a PDF-to-practice-question workflow that extracts and chunks study material, then uses a LangChain refinement chain and Gemini to generate a focused set of review questions.
- Created a retrieval question-answering pipeline by embedding document chunks with Ollama and indexing them in FAISS, allowing learners to reveal answers grounded in their uploaded source.
- Developed a responsive FastAPI and Jinja2 interface with PDF upload validation, generation feedback, interactive question cards, on-demand answer reveal, and one-click question copying.

AI Coding Helper
Python, FastAPI, LangGraph, LangChain, PostgreSQL, pgvector
- Developed an AI coding mentor and review assistant that helps developers reason through problems with progressive hints, explanations, and guiding questions.
- Orchestrated a stateful LangGraph workflow covering input guardrails, document processing, context retrieval, response generation, output validation, and conversation summarization.
- Enabled semantic search over uploaded code using LlamaIndex CodeSplitter, Sentence Transformers, PostgreSQL, and pgvector; added user-specific context with mem0 and Redis/Valkey caching.
- Connected authentication, asynchronous APIs, SSE streaming, and rate limiting, with Langfuse tracing, Prometheus metrics, Grafana dashboards, Docker packaging, and MCP tools.

GNSS Spoofing Detection System
Python, scikit-learn, Signal Processing
- Built a multi-class detector for seven GNSS spoofing scenarios, achieving 99.6% accuracy and a 0.99 macro F1-score on a held-out set of 485 samples.
- Engineered signal features from labeled data and benchmarked Logistic Regression, Random Forest, SVM, and neural-network models with per-class precision, recall, F1-score, and accuracy.

Real-Time Object Detection and Segmentation | *GitHub*
Python, PyTorch, YOLO, SAM, OpenCV, CUDA, Streamlit, Docker
- Combined YOLO object detection with SAM pixel-level mask generation to create an end-to-end image and video analysis workflow with these two technologies.
- Built an interactive Streamlit interface for visualizing detections and segmentation overlays, and used CUDA acceleration for GPU inference.
- Packaged the application with Docker to make environment setup repeatable and simplify deployment across compatible machines.

Malware Detection System | *GitHub*
Python, scikit-learn, Pandas, NumPy
- Developed a malware classifier reported at 97% accuracy by extracting static and behavioral features and comparing Random Forest, SVM, and neural-network approaches.
- Added USB monitoring to launch scans when a device is connected, and evaluated model performance with precision, recall, F1-score, and confusion matrices.

Experience

Sprints **Jul. 2026 – Aug. 2026**
AI Engineer Intern
- Built an AI coding mentor with LangGraph and FastAPI, organizing code analysis, document processing, context retrieval, hint generation, response validation, and conversation summaries into a stateful workflow.
- Implemented semantic search over uploaded code using LlamaIndex CodeSplitter, Sentence Transformers, PostgreSQL, and pgvector to retrieve relevant code context for each request.
- Added personalized coaching with mem0 long-term memory, skill-profile generation, and Redis/Valkey caching to reuse useful user context across interactions.
- Integrated asynchronous APIs, JWT authentication, SSE streaming, rate limiting, and database persistence to support secure and responsive application workflows.
- Used Docker for packaging and connected Langfuse tracing, Prometheus metrics, Grafana dashboards, and MCP tools to support observability and tool integration.

Digital Egyptian Pioneers Initiative (DEPI) **Jun. 2025 – Jan. 2026**
Data Engineering Trainee
- Practiced collecting, cleaning, transforming, and storing data through applied exercises and projects using relational datasets.
- Wrote SQL queries for data extraction and analysis, and worked through ETL concepts covering ingestion, transformation, and loading.
- Studied data modeling, data warehousing, and big-data fundamentals, with an introduction to cloud data solutions and modern pipeline workflows.

Talaat Mostafa Group (TMG) **Jul. 2025 – Aug. 2025**
IT Intern *Alexandria, Egypt*
- Diagnosed hardware and software issues across five workstations and helped improve system uptime while reducing the support-ticket backlog.
- Configured Windows environments for new users, including accounts, network settings, and security policies; troubleshot MySQL access and supported routine database backups.
- Assisted with antivirus updates, firewall-log monitoring, and basic phishing-prevention guidance as part of the team's cybersecurity support.

Abu Qir Petroleum **Aug. 2024 – Sep. 2024**
IT Intern
- Gained on-site exposure to enterprise server structures, server types, and the facilities used to maintain production IT infrastructure.
- Studied network programming fundamentals and how connected devices exchange information within an organization's network environment.

Information Technology Institute (ITI) **Jul. 2023 – Oct. 2023**
Java Development Trainee
- Completed a 120-hour Java program covering object-oriented programming, data structures, and modular application design.
- Built a library-management GUI supporting book borrowing, return tracking, overdue detection, and early-return handling; the project was evaluated by ITI instructors and graded A+.

Certifications

AI and Data Science – Microsoft Data Engineer | ***Certificate*** **Jun. 2025 – Jan. 2026**
Digital Egyptian Pioneers Initiative

DEPI English Course | ***Certificate*** **Dec. 2025 – Jan. 2026**
Digital Egyptian Pioneers Initiative

TMG IT Training | ***Certificate*** **Jul. 2025 – Aug. 2025**
Talaat Mostafa Group

Introduction to Artificial Intelligence | ***Certificate*** **Feb. 2025**
Ministry of Communications and Information Technology

NU Quantum Hackathon | ***Certificate*** **Jul. 2024**
Nile University

Leadership and Volunteering

E-JUST Organization Team **Jul. 2023 – Present**
Organizer *Egypt-Japan University of Science and Technology*
- Supported planning and on-site coordination for entrance examinations and admissions events serving more than 5,000 applicants across four admission cycles.
- Coordinated logistics with multiple university faculties and contributed to large events attended by official representatives of the Japanese government.